

Track 1 — Methods description

Team: ciberpty · MVA Hackathon 2026 · Proband PROBAND01
Repository: <https://github.com/cpu-16/mva-hackathon-2026>

Q3 — Describe your model/approach in detail

We ran two independent analyses and required them to converge before accepting a result.

Analysis A — genome-wide, phenotype-informed, hypothesis-free (primary)

Exomiser 15.1.0 with the 2602 hg38 and 2602 phenotype data releases. Input was the unmodified proband VCF — **5,012,204 records, of which 4,740,790 carry the PASS filter** (GRCh38, GATK 4.2.4.0) — together with the eight HPO terms extracted verbatim from the supplied clinical document:

HPO	Term
HP:0002859	Rhabdomyosarcoma
HP:0000121	Nephrocalcinosis
HP:0004322	Short stature
HP:0001508	Failure to thrive
HP:0003202	Skeletal muscle atrophy
HP:0001622	Premature birth
HP:0001518	Small for gestational age
HP:0200067	Recurrent spontaneous abortion

We note that the last term describes the parents' obstetric history rather than a finding in the child. We kept it because the clinical document states it and because it is informative for the mode of inheritance, but we flag it explicitly rather than let it pass as a proband phenotype.

No gene panel, no candidate list and no reference to a disease name entered the configuration at any point. Prioritisation used `hiPhivePrioritiser` (human, mouse, fish and protein-interaction evidence) and `omimPrioritiser`. Inheritance models covered autosomal dominant, autosomal recessive (homozygous and compound heterozygous), X-linked and mitochondrial. Frequency sources were UK10K and gnomAD exomes/genomes across five populations at a 2% maximum; pathogenicity sources were REVEL, MVP, AlphaMissense and SpliceAI.

Filtering cascade: Exomiser loaded 4,962,047 of those records; 4,702,177 passed the failed-variant filter → 277,724 after the variant-effect filter → 11,255 after frequency → 9,935 after inheritance-mode compatibility, distributed across 3,139 genes.

Result

BUB1B ranked 2nd of 3,139 genes under the autosomal recessive model, and Exomiser assigned that ranking to ORPHA:1052, mosaic variegated aneuploidy syndrome — the correct disease, reached without being told. Under the recessive model both variants were marked as contributing, so Exomiser reconstructed the compound heterozygous pair on its own.

The same gene also took rank 1 under the autosomal dominant model, but that entry is an artefact we report rather than claim: Exomiser attached it to **OMIM:114500, colorectal cancer (somatic)**, and it wins the top slot largely because the nonsense variant is on the ClinVar whitelist, which short-circuits variant scoring to 1.0. A single truncating allele in a recessive gene is not a dominant colorectal cancer diagnosis. **The result we stand behind is rank 2, not rank 1.** We say so because the honest reading of our own output matters more than the better-sounding headline.

The combined score gap is real but moderate: 0.5871 (rank 1, AD) and 0.5538 (rank 2, AR) against 0.2137 for the third-ranked gene — **2.59× the third gene for the recessive result we stand behind** (2.75× for the dominant artefact), not an order of magnitude.

Robustness controls — including one that argues against us

We ran eleven additional Exomiser analyses to test how much of the result comes from the phenotype and how much from the variant.

Leave-one-HPO-out (8 runs). Removing each HPO term in turn: **BUB1B remained rank 1 in all eight**, with scores from 0.5447 to 0.7503. Removing the rhabdomyosarcoma term — the one a human would consider most diagnostic — actually *raised* the score to 0.7503. No single symptom carries the result.

Unrelated HPO terms (1 run). Replacing the eight clinical terms with five clinically unrelated ones — seizure (HP:0001250), hearing impairment (HP:0000365), atrial septal defect (HP:0001631), conjunctivitis (HP:0000509) and recurrent respiratory infections (HP:0002205) — **BUB1B still came out rank 1**, score 0.5506.

Least-specific term alone (1 run). With short stature (HP:0004322) as the only input, BUB1B fell to **rank 4** and HK1 took the top slot.

No phenotype at all (1 run). Exomiser terminates with an error; the tool requires at least one HPO term, so this control could not be run.

What this forces us to conclude. We cannot call this a phenotype-driven discovery. The signal comes from the **variant** — a ClinVar-whitelisted pathogenic nonsense allele in a constrained gene (LOEUF 0.588) — not from the phenotype. The accurate description is **variant-driven and phenotype-consistent**. What the phenotype does contribute is the choice between competing interpretations of the same variant: it is the phenotype that makes Exomiser attach the recessive model to mosaic variegated aneuploidy rather than to somatic colorectal cancer, and the least-specific-term control shows the effect is real (one vague term is not enough to reach the top). That is a narrower claim than “our pipeline found it from the symptoms”, and it is the one the data support.

A twelfth run removed the variant-effect filter entirely (219,399 variants across 13,338 genes, 22× and 4× the primary run) and returned identical scores and an unchanged top ten. We report it as a

technical check that the filter did not manufacture the result, not as evidence of robustness — the causal variants are coding, so the filter was never going to remove them.

Analysis B — targeted panel, orthogonal confirmation

An 11-gene panel of MVA genes and phenocopies (BUB1B, CEP57, TRIP13, CENATAC, MAD1L1, MAD2L1BP, CEP192, BUB1, SMC5, TRIM37, CENPE) with 5 kb padding. Coordinates were retrieved programmatically from the **Ensembl REST API (release 116)** rather than from memory or a static file, and the provenance of every coordinate is recorded. The 1,532 PASS variants inside those intervals were annotated with the **Ensembl VEP REST API** and filtered for HIGH or MODERATE impact on the canonical transcript at gnomAD AF below 1% or absent.

Exactly three variants survived. Two were in BUB1B; the third (MAD1L1 p.Arg59Cys) was a single allele in a recessive gene and cannot explain the phenotype. **BUB1B was the only gene carrying two rare damaging alleles.**

Convergent answer

Variant	Consequence	gnomAD (max pop. AF)	ClinVar	Exomiser ACMG (AR model)
NM_001211.6:c.2210T>G p.Leu737Ter	stop_gained	9.98×10 ⁻⁵ (gnomAD exomes NFE)	533901, Pathogenic/ Likely pathogenic, 2★	PATHOGENIC (PVS1, PM2_Supporting, PP4_Moderate, PP5_Strong)
NM_001211.6:c.3006T>G p.Asn1002Lys	missense	8.99×10 ⁻⁷ (gnomAD exomes NFE)	not present	UNCERTAIN_SIGNIFICANCE (PM2_Supporting, PP4_Moderate, BP1)

Two notes on this table, both of which correct looser statements we could have made:

- **The two frequency sources disagree for the nonsense variant.**
Exomiser's bundled gnomAD gives 9.98×10⁻⁵ (GNOMAD_E_NFE); the Ensembl VEP REST API returned $\approx 3 \times 10^{-5}$ for the same variant. We report the Exomiser figure because it is the one the ranking

was computed from, and we declare the discrepancy rather than pick the smaller number. Either way the allele is rare and there are no homozygotes.

- **The missense variant is not absent from gnomAD**, as we first wrote. It is present at 8.99×10^{-7} , which is a single-observation-level frequency but not zero. It carries REVEL 0.472, MVP 0.852 and AlphaMissense 0.923 — the in-silico predictors disagree with each other, which is why we let it stand as a VUS rather than upgrade it.

On ClinVar: **c.3006T>G is not in ClinVar**. ClinVar contains `c.3006T>A` (VCV004600147, `NC_000015.10:40220611:T:A`), a *different nucleotide substitution at the same position* producing the same p.Asn1002Lys protein change, classified Uncertain significance — criteria provided, single submitter, last evaluated 2025-09-19. That classification does **not** transfer: the two alleles differ in splicing potential and in mutational context, and a VUS transferred across alleles would in any case add no evidence. We did not use it.

Interpretation: findings consistent with mosaic variegated aneuploidy type 1 (MVA1, OMIM 257300), autosomal recessive, through compound heterozygosity in BUB1B (phase not demonstrated; see limitations). The truncating-plus-missense configuration matches the allelic pattern described in biallelic MVA patients — “in patients with biallelic mutations, a missense mutation pairs with a truncating mutation” (Suijkerbuijk 2010, PMID 20516114). Reading that pair as null-plus-hypomorph is our inference from the same work’s observation of absent transcript from truncating alleles and rapid turnover of missense product; the source does not state that complete biallelic loss is lethal, and neither do we.

Additional analysis — mosaicism (negative result, with a computed detection limit)

Because MVA is defined by mosaic aneuploidy, we tested whether that signature is visible in the bulk WGS: per-chromosome B-allele frequency across 2.27 M heterozygous SNVs; the same restricted to a fixed depth window (DP 40–48) to remove the coverage confounder; a spatial profile in 5 Mb windows; and chromosomal dosage from 10%-trimmed mean depth. Three of these are refinements of the same BAF

statistic and only the dosage test is orthogonal — they are not four independent tests. Only the VCF was available, with no BAM, so no GC-LOESS normalisation was possible.

We computed the detection limit instead of asserting a negative, and we state where the threshold comes from. Across the 22 autosomes at fixed depth (DP 40–48, 857,435 sites), the measured standard deviation of mean $|BAF - 0.5|$ between chromosomes is **0.0035**. Our 0.005 noise floor is therefore a ≈ 1.4 -SD criterion. Simulating mean $|BAF - 0.5|$ at DP 44 against it, a clonal trisomy becomes detectable only when present in **$\sim 14\%$ of cells**.

The conclusion does not depend on that choice, and every stricter choice hurts the assay rather than us: a conventional 2-SD criterion moves the limit to **16%**, and 3-SD to **20%**.

Read sampling is not what sets that limit. The simulation subtracts the binomial baseline at DP 44, so the 0.005 threshold a signal must clear is the *measured dispersion between chromosomes* — systematic library bias, not counting noise. The read-depth dosage test points the same way by a separate route: Poisson error when averaging a whole chromosome at $44\times$ is $\approx 0.03\%$, orders of magnitude below the deviations observed. These are two different statistics and we do not conflate them; they agree on the conclusion that **more sequencing depth would not move the limit**.

Every candidate signal traced to coverage, segmental duplications (22q11, centromeres) or — partially — GC content. We flag that the GC explanation is incomplete: the measured GC-coverage correlation was $r = 0.43$ from a sparse sample, and chr21, the largest deviation, is GC-poor. Where dosage and BAF disagreed we favoured BAF, which is an internal per-site ratio and cancels coverage bias, and we report the dosage result as an open question.

We therefore do not claim the child lacks mosaic aneuploidy — he has it by diagnostic definition. We claim this assay cannot resolve it. Variegation spreads the burden across chromosomes: an aneuploid fraction of $\sim 30\%$ distributed over the autosomes leaves each individual chromosome altered in roughly **1–2% of cells, which is 7–16 $\times$ below the 14–16% detection limit**. Bulk averaging erases variegation by construction, at any depth. Premature chromatid separation, the

cytogenetic hallmark of MVA, leaves no trace in DNA sequence at all. The practical consequence is that **aneuploidy burden is not a measurable endpoint in these data**, which constrains any Track 2 proposal whose outcome depends on it.

Q4 — Automated output or manual review?

Automated output with documented manual review. The submitted CSV is the direct result of the convergence between the two analyses, but two decisions were made by hand — confirming the compound-heterozygous pair and adjudicating the secondary candidates — and reporting the submission as purely automated would be inaccurate.

Q5 — Describe the manual review

Manual review was limited to two decisions, both documented:

1. **Confirming the compound-heterozygous pair.** Both analyses proposed the pair automatically; we verified the genotypes (0/1 for both), read depths (46× and 28×) and allelic balance (21,25 and 15,13) directly in the VCF.
2. **Adjudicating two secondary candidates** (see Q10).

No variant was added, removed or re-ranked on intuition.

Q6 / Q7 — Data sources

Public data only. No proprietary data sources were used.

- Ensembl REST API release 116 (gene coordinates)
- Ensembl VEP REST API (functional consequence, canonical transcript, gnomAD frequencies)
- Exomiser 15.1.0 with data releases 2602_hg38 and 2602_phenotype
- gnomAD v4 (via Exomiser and VEP)
- ClinVar, GRCh38 (NCBI FTP snapshot and E-utilities)
- Human Phenotype Ontology (`hp.obo` , `genes_to_phenotype.txt`)
- REVEL, MVP, AlphaMissense and SpliceAI (via the Exomiser bundle)
- OMIM and Orphanet identifiers via the Exomiser `omimPrioritiser`

- Published literature (PubMed / Europe PMC), cited by PMID

Q9 — Can the approach output compound heterozygous pairs?

Yes, and it did so without being told to. Exomiser's `AUTOSOMAL_RECESSIVE_COMP_HET` mode flagged both BUB1B variants as contributing under the recessive model. The targeted analysis independently selected any gene carrying two rare damaging alleles, which returned BUB1B alone.

Neither method can establish phase, and the data show why. There is no parental sample, and the two variants lie ~11 kb apart, far beyond read-backed phasing range for Illumina paired-end reads. This is not a theoretical limit in this VCF: GATK emitted no `PGT / PID` phasing tags for either BUB1B variant, while it *did* phase three FANCD2 variants spanning 5 bp into a single haplotype (`PID=10046720_C_T`) elsewhere in the same file. Read-backed phasing worked where the physical distance allowed it and was silent where it did not.

The compound-heterozygous interpretation therefore rests on allele rarity, the known recessive inheritance of MVA1, the phenotype match, and the absence of homozygotes in gnomAD — not on a demonstrated *trans* configuration. Parental testing would settle it, and it is the single highest-value follow-up experiment for this case.

Q10 — Secondary / incidental findings

None reported. The Exomiser output contains 12,431 variant-by-inheritance-model rows, 5,348 of which carry an ACMG classification. **Three unique variants** reached pathogenic or likely pathogenic (four rows — the BUB1B nonsense appears twice, once under the AD model and once under the AR model). Two of the three lie outside BUB1B, and both were adjudicated as non-reportable:

FANCD2 chr3:10046723 AG>A — ENST00000675286.1:c.1278+1del, Exomiser ACMG PATHOGENIC. Three independent reasons not to report it, in order of weight:

1. **The variant is not isolated: three co-phased changes in one splice region.** GATK phased it onto a single haplotype

(PID=10046720_C_T) with two neighbours within 6 bp — chr3:10046720 C>T, chr3:10046723 AG>A and chr3:10046725 TAAG>T (c.1278+3_1278+5del). All three sit at an allelic fraction of ≈ 0.31 (AD 65,31 / 61,28 / 66,28) rather than the ≈ 0.5 expected of a germline heterozygote. Three clustered, co-phased changes at a consistent sub-heterozygous fraction is the signature of a divergent or mismapped haplotype, not of three independent events. (An earlier draft described the second variant as an SNV 2 bp away; it is in fact a 3 bp deletion, which strengthens the mismapping reading rather than weakening it.)

2. **FANCD2 is a known short-read mapping pitfall.** The gene has high-identity pseudogenes, and the Fanconi anaemia sequencing literature documents that “database errors and in particular pseudogenes impose obstacles that may prevent correct data perception and interpretation” (PMID 23285130).
3. **ClinVar disagrees with Exomiser, though we weight this last.** ClinVar VariationID 3059927 classifies the deletion **Benign**, while Exomiser calls it **PATHOGENIC** because its PVS1 rule fires on any canonical splice-donor deletion with no ClinVar downgrade applied. We put this third rather than first because the ClinVar record is weak evidence in its own right: **0 stars, no assertion criteria provided, a single submitter** (PreventionGenetics, SCV004798965.2). The neighbouring deletion (ClinVar 518343, Benign/Likely benign) carries the same limitation. A 0-star benign call does not by itself refute a pathogenic prediction — it is the co-phased haplotype and the allelic fraction that do the work here.

Even if the deletion were real, a single allele of an autosomal recessive disorder is carrier status. Sanger sequencing or long reads would be needed before any clinical report.

GNRHR chr4:67753920 C>T — c.416G>A p.Arg139His, Exomiser ACMG LIKELY_PATHOGENIC. A clean heterozygote (AD 28,32, DP 60) and ClinVar Pathogenic (VariationID 16030, multiple submitters, no conflicts) for hypogonadotropic hypogonadism 7 with or without anosmia (OMIM 146110) — an **autosomal recessive** condition, so a single allele is carrier status. The allele reaches 3.3% in gnomAD Latino/Admixed American, consistent with a recessive carrier allele rather than a dominant finding, and GNRHR does not appear on the

ACMG SF v3.2 secondary findings gene list (statement: PMID 37347242; gene table as reproduced by NCBI ClinVar, checked directly). Carrier status is not a reportable secondary finding, and the condition is unrelated to this child's phenotype.

We chose to document the reasoning rather than submit these as `secondary` rows. They would not have harmed the automated score, but reporting them as findings would be a clinically meaningful false positive.

Q11 — Runtime and cost

- Exomiser genome-wide run: **53 seconds** wall clock (32-core CPU, 16 GB JVM heap).
- Eleven robustness controls: about 10 minutes total.
- Targeted panel with REST annotation: under 5 minutes, dominated by API latency.
- Mosaicism analysis: about 10 minutes of `bcftools` streaming.
- One-off setup: $\approx$ 55 GB of Exomiser reference data downloaded and extracted.
- **Marginal compute cost is effectively zero** — a commodity workstation, no GPU, no cloud instance. Reproducing this on a laptop is realistic: only the 315 MB VCF and the phenotype document are needed, not the 85 GB of raw reads.

Q12 — Method abstract ($\leq$ 500 words)

We identified the causal variants through two independent analyses designed to check each other, and required them to converge before accepting a result.

The primary analysis was deliberately hypothesis-free. Exomiser 15.1.0 (data release 2602, hg38) was run on the complete proband VCF — 4,740,790 PASS variants — driven only by the eight HPO terms taken verbatim from the clinical document. No gene panel, no candidate list and no mention of a disease name entered the configuration. Prioritisation combined hiPhive (human, mouse, fish and protein-interaction evidence) with the OMIM prioritiser across dominant, recessive, X-linked and mitochondrial models. **BUB1B ranked 2nd of**

3,139 genes under the recessive model, mapped by Exomiser to mosaic variegated aneuploidy syndrome (ORPHA:1052), with both variants flagged as contributing — the compound heterozygous pair reconstructed without instruction. The same gene took rank 1 under the dominant model, but attached to somatic colorectal cancer; we report that entry as an artefact of the ClinVar whitelist rather than claim it.

The confirmatory analysis took the opposite approach: an 11-gene panel of MVA genes and phenocopies, coordinates fetched programmatically from the Ensembl REST API with provenance recorded, annotated through Ensembl VEP and filtered for HIGH/MODERATE impact below 1% gnomAD AF. Of 1,532 variants three survived; two were in BUB1B, the only gene carrying two rare damaging alleles.

Both routes converge on **NM_001211.6:c.2210T>G p.Leu737Ter** (nonsense; ClinVar 533901 Pathogenic/Likely pathogenic; gnomAD 9.98×10^{-5}) and **c.3006T>G p.Asn1002Lys** (missense; gnomAD 8.99×10^{-7} ; not in ClinVar), consistent with MVA1 (OMIM 257300) and with the truncating-plus-missense allelic pattern of viable MVA1.

Eleven robustness controls, one of which argues against us. Leave-one-HPO-out keeps BUB1B at rank 1 in all eight runs, so no single symptom drives the result — but substituting clinically unrelated HPO terms *also* returns BUB1B at rank 1. We therefore describe the finding as **variant-driven and phenotype-consistent**, not phenotype-driven: the ClinVar-whitelisted nonsense allele carries the ranking, and the phenotype's real contribution is selecting the recessive MVA interpretation over the dominant cancer artefact. Reporting this control costs us an argument; omitting it would have been the kind of thing a reviewer finds.

We also asked whether mosaic aneuploidy is visible in the bulk WGS. It is not, and we computed why: simulating mean $|BAF - 0.5|$ at DP 44 puts the detection limit at ~14% of cells for a clonal trisomy (16% under a stricter 2-SD criterion), while variegation leaves each chromosome at 1–2% of cells — 7–16× below it. Aneuploidy burden is not a measurable endpoint in these data — a constraint that matters for evaluating any candidate therapy.

Limitations. Phase cannot be established: no parental sample, and the variants lie ~11 kb apart, beyond read-backed phasing (confirmed by the absence of GATK phase tags on both). The functional consequence of p.Asn1002Lys is unproven, and its in-silico predictors disagree (REVEL 0.472, AlphaMissense 0.923). REMM and CADD were omitted. The genome-wide Exomiser run took 53 seconds on a commodity workstation with no GPU.

Q23 — Generative AI declaration

Commercially-available generative AI was used. Provider, plan and relevant setting for each:

- **Anthropic**, Claude Opus via Claude Code, **Max** subscription plan, model-improvement data sharing disabled in account privacy settings.
- **OpenAI**, Codex (gpt-5.6), **Plus** plan, “Improve the model for everyone” disabled.
- **xAI Grok 4.6 via Cursor**, **Pro** plan, **Privacy Mode enabled** (zero data retention at Cursor and at the underlying model providers).
- **Piper TTS** (`en_US-ryan-high`) — open-source, run entirely on local hardware with no cloud service and no API key. Used only to narrate the pitch video. Listed for completeness; it is not a commercial service.

How these tools were used, and how they were not. The language models were used for literature retrieval and citation checking, for adversarial review of our own drafts, and for writing. They were also used against us on purpose: two independent models were asked to attack our reports, and the corrections they produced — including a claim of ours that contradicted our own data file — are recorded in the repository under `evidencia/`.

No block of the VCF and no genotype table was ever sent to a language model. All genomic work ran on local tools (`bcftools`, Exomiser 15.1.0) and on annotation APIs — Ensembl VEP and ClinVar — that the hackathon policy names explicitly as acceptable. What the models saw were named variants, HPO terms and reasoning: exactly the category the policy classifies as a “finding” and permits retaining.